

United States Spotify Top 50

Playlist Performance and Song Popularity Trend Analysis

Research Paper

Field	Detail
Prepared For	Atlantic Recording Corporation
Analyst	Himanshu Yadav
Dataset Period	May 18, 2024 – November 27, 2025
Total Records	27,752 daily playlist snapshots
Unique Songs	938 tracks across 296 artists
Dashboard	spotify-us-top50.streamlit.app

1. Introduction

Music consumption in the streaming era is highly dynamic. Song popularity changes daily, playlist rankings directly influence streams and revenue, and artist visibility depends on chart persistence rather than peak rank alone. For record labels like Atlantic Recording Corporation, understanding playlist behavior over time is crucial for making informed decisions about artist promotion strategies, release timing, marketing spend optimization, and catalog performance analysis.

This research paper presents a comprehensive analysis of the US Spotify Top 50 playlist spanning 558 days from May 18, 2024, to November 27, 2025. The dataset comprises 27,752 daily playlist snapshots tracking 938 unique songs across 296 artists, capturing chart position, popularity scores, audio attributes, and content metadata.

This project focuses exclusively on historical playlist analytics not prediction or recommendation systems. The goal is data-driven business intelligence: understanding what drives sustained chart success, who dominates the playlist and how, and what content attributes correlate with chart longevity.

1.1 Project Objectives

- Validate and analyze the structure and quality of the daily playlist dataset
- Engineer meaningful performance KPIs: Days on Chart, Rank Volatility Index, Popularity Trend Score, and Artist Dominance Index
- Identify patterns in ranking stability, artist dominance, and content attribute performance
- Deliver actionable recommendations for Atlantic Recording Corporation's promotion and release strategy
- Build a live, publicly accessible analytics dashboard at spotify-us-top50.streamlit.app

1.2 Research Questions

- How strongly does Spotify popularity score correlate with chart rank?
- Which artists dominate the US Top 50 and what strategies drive their dominance?
- Does explicit content perform differently from clean content in terms of rank and longevity?
- What song attributes (duration, album type) are associated with chart persistence?
- What is the typical chart lifecycle of a song, and what separates long-tail hits from one-week wonders?

2. Dataset Description

The dataset consists of daily snapshots of the US Spotify Top 50 playlist, sourced via the Atlantic Recording Corporation API. Each row represents a single song's presence on the playlist on a given date, capturing its rank, audio metadata, and popularity score.

2.1 Dataset Overview

Attribute	Value
Total rows	27,752
Total columns	10 (+ 8 engineered features)
Date range	May 18, 2024 – November 27, 2025
Days tracked	558
Unique songs	938
Unique artists	296
Avg songs/day	50 (full playlist coverage)
Avg popularity	87.7 / 100

2.2 Column Descriptions

Column	Type	Description
date	datetime	Date of the playlist snapshot
position	integer	Chart rank (1 = highest, 50 = lowest)
song	string	Song title
artist	string	Primary artist name(s)
popularity	integer	Popularity score from Atlantic API (0–100)
duration_ms	integer	Song duration in milliseconds
album_type	string	Album classification: Single, Album, or Compilation
total_tracks	integer	Total number of tracks in the parent album
is_explicit	boolean	Explicit content flag (True/False)
album_cover_url	string	URL of the album artwork image

3. Data Cleaning & Validation

All data was processed through a structured preprocessing pipeline before analysis. The pipeline applied standardized validation checks, type enforcement, and normalization steps to ensure analytical integrity.

3.1 Validation Results

Validation Check	Result	Status
Missing required columns	0 missing	PASS
Invalid rank values (outside 1-50)	0 rows	PASS
Duplicate song-date-artist rows	0 duplicates	PASS
Null values (all columns)	0 nulls	PASS
Unparseable date entries	0 rows	PASS

The dataset passed all validation checks without exception. This is a noteworthy result, real-world music datasets typically suffer from artist name inconsistencies (e.g., featuring artist variations), duplicate entries from API pagination errors, and date format inconsistencies. The absence of all these issues indicates a well-maintained data source and allows full confidence in downstream analysis without statistical bias from imputation.

3.2 Cleaning Steps Applied

- Text normalization: song and artist names stripped of leading/trailing whitespace; lowercase key columns created for safe groupby operations without casing-related duplicate groups
- Date parsing: converted from raw string to datetime64 using pandas.to_datetime() with error coercion
- Type enforcement: position and total_tracks cast to nullable Int64; is_explicit mapped from True/False strings to boolean
- Album type normalization: title-cased to ensure consistent groupby (e.g., 'single' and 'Single' unified)
- Rank filtering: any rows with position outside 1–50 would be dropped (none found in this dataset)
- Deduplication: exact duplicate rows on song+date+artist removed (none found)
- Sort: final DataFrame sorted by date then position for correct time-series ordering

4. Feature Engineering

Eight derived features were engineered from the raw data to enable the KPI analysis specified in the project requirements. All features are computed using vectorized Pandas operations, no Python loops, ensuring performance efficiency.

4.1 Song-Level KPIs

KPI	Formula	Purpose
Days on Chart	<code>groupby(song).date.nunique()</code>	Chart longevity indicator
Average Rank	<code>groupby(song).position.mean()</code>	Overall chart performance
Best Rank	<code>groupby(song).position.min()</code>	Peak achievement metric
Rank Volatility Index	<code>groupby(song).position.std()</code>	Stability (NaN filled to 0)
Avg Popularity	<code>groupby(song).popularity.mean()</code>	Listener engagement score
Duration (minutes)	<code>duration_ms / 60,000</code>	Standardized duration
Popularity Trend Score	<code>np.polyfit slope of popularity vs time</code>	Rising vs fading trajectory
Rank Change	<code>groupby(song).position.diff()</code>	Daily rank movement

4.2 Artist-Level KPIs

KPI	Formula	Purpose
Total Appearances	<code>groupby(artist).date.count()</code>	Presence volume
Unique Songs	<code>groupby(artist).song.nunique()</code>	Catalog breadth
Artist Avg Rank	<code>groupby(artist).position.mean()</code>	Consistent performance
Dominance Index	<code>artist_appearances / total_playlist_slots x 100</code>	Market share %

4.3 Popularity Trend Score Methodology

The Popularity Trend Score is computed as the linear slope of a song's popularity over time using `numpy.polyfit()`. A positive slope indicates rising listener engagement; a negative slope indicates a fading song. Songs with fewer than 2 appearances receive a score of 0. This metric was used to identify chart risers and chart decliners in the Streamlit dashboard's Song Performance module.

5. Exploratory Data Analysis

5.1 Distribution of Core Variables

Popularity scores are left-skewed and concentrated between 80–100, with a mean of 87.7. The compressed range means popularity provides limited discriminatory power between songs on this playlist, virtually all songs have high popularity, which helps explain the weak correlation with rank explored in Section 5.2.

Chart position is roughly uniformly distributed across all 50 slots, as expected for a fixed-size daily playlist. There is slightly higher frequency at positions 40–50, indicating greater churn at the bottom of the chart than at the top.

Song duration peaks in the 2–4 minute range, consistent with streaming-era production conventions. Very few songs exceed 5 minutes. The near-total absence of very short or very long songs on this playlist suggests moderate-length tracks are preferred by the algorithmic and editorial curation process.

The days-on-chart distribution is strongly right-skewed. The majority of songs appear for fewer than 30 days, while a small elite group persists for 300–500+ days. This J-curve pattern is universal in hit music markets: a flash-in-the-pan majority and a perennial minority.

5.2 Popularity vs. Chart Rank Correlation

One of the most strategically important findings in this analysis is the near-zero correlation between Spotify popularity score and chart rank.

Statistic	Value	Interpretation
Pearson r	-0.094	Very weak negative correlation
R-squared	0.0099	Popularity explains only ~1% of rank variance
OLS slope	-0.075	Near-flat relationship
OLS intercept	89.47	Expected popularity at rank 1 ~89.5

***Chart rank and popularity score measure fundamentally different things.
Popularity reflects rolling global streaming activity; rank reflects daily domestic
streaming velocity on this specific playlist.***

A song can be globally popular but not trending in the US on a given day, and vice versa. This divergence explains why the correlation is near zero despite the intuitive expectation that popular songs should rank highly. Teams at Atlantic should treat these as independent metrics requiring separate monitoring strategies.

5.3 Chart Longevity Distribution

Percentile	Days on Chart	Interpretation
25th percentile	3 days	Bottom quarter, brief appearances
50th (median)	12 days	Half of all songs disappear within 2 weeks
75th percentile	38 days	Top quarter, notable performers
90th percentile	95 days	Strong sustained presence
95th percentile	155 days	Elite long-tail assets
Maximum	536 days	Something in the Orange by Zach Bryan

The median song lasts only 12 days on the chart. The mean of 27 days is substantially inflated by outliers like Something in the Orange (536 days). This stark divide reinforces the business case for identifying and doubling down on songs in their first 30 days, this is the critical intervention window.

6. Key Findings

6.1 Artist Dominance

The US Top 50 is highly concentrated at the top. The three most dominant artists are Taylor Swift, Zach Bryan, and Morgan Wallen, collectively account for approximately 20% of all playlist slots across the 558-day period.

Rank	Artist	Appearances	Unique Songs	Dominance %	Strategy
1	Taylor Swift	2,015	96	7.26%	Catalog Breadth
2	Zach Bryan	1,857	35	6.69%	Album Cut Depth
3	Morgan Wallen	1,669	10	6.01%	Hit Concentration
4	Sabrina Carpenter	967	14	3.48%	Rising Crossover
5	Drake	841	35	3.03%	Catalog Depth
6	Chappell Roan	794	6	2.86%	Breakout Artist
7	Travis Scott	762	31	2.75%	Album Release Cycle
8	Billie Eilish	729	12	2.63%	Global Crossover

Three distinct dominance strategies emerge from the data. First, Taylor Swift's Catalog Breadth model (96 songs) leverages an entire discography every tour, re-release, and cultural moment pulls old catalog tracks back into the chart. This is the highest-ceiling strategy but requires years of catalog accumulation. Second, Zach Bryan's Album Cut Depth model (35 songs) reflects strong album-oriented listeners who stream multiple tracks from one release simultaneously. Third, Morgan Wallen's Hit Concentration model (10 songs, 1,669 appearances) demonstrates that a handful of inescapable hits can match catalog-heavy competitors in total dominance with the highest streams-per-song efficiency of any top artist.

6.2 Explicit Content Performance

Metric	Clean Songs	Explicit Songs	Difference
Unique songs on chart	457	496	Explicit: +8.5% more titles
Average chart rank	25.60	25.40	Explicit: marginally better
Average popularity	87.99	87.30	Clean: +0.69 points
Average days on chart	163.3	135.3	Clean: +28 days (+20.7%)

Clean songs stay on the chart 28 days longer than explicit songs on average a 20.7% longevity advantage that translates directly to sustained streaming revenue.

Despite explicit content having a marginal rank advantage, clean tracks demonstrate significantly greater chart persistence. The likely mechanism is discovery surface: clean tracks are eligible for family playlists, all-ages editorial features, and radio playlisting all of which extend the streaming tail beyond the initial chart cycle.

6.3 The Longest-Charting Song

Something in the Orange by Zach Bryan charted for 536 of the 558 days tracked a near-perfect attendance record. This song exemplifies what separates a cultural anchor from a typical chart entry: it combines Zach Bryan's loyal album-listening fanbase with a song that has become synonymous with a specific emotional moment in popular culture, generating consistent organic daily streams that maintain chart position without relying on promotional push.

6.4 Content Attribute Summary

Attribute	Finding	Business Implication
Explicit vs Clean	Clean songs chart 28 days longer	Always release a clean version
Album vs Single	Singles dominate chart presence	Singles-first release strategy
Duration sweet spot	2–4 minutes achieves highest avg popularity	Target standard pop song length
Popularity vs Rank	$r = -0.094$ (near zero correlation)	Track US rank independently
Top 3 artist share	~20% of all playlist slots	Sustained investment required to compete

7. Business Recommendations

7.1 Release Strategy

- Always release a clean version of every track simultaneously with the explicit version. The +28 day longevity advantage for clean content translates directly to 4+ additional weeks of streaming revenue per release.
- Prioritize singles-first release cadence for new Atlantic artists. Album tracks underperform singles on this playlist in both chart frequency and average days on chart.
- Target the 2–4 minute song duration range for tracks intended for chart placement. This aligns with both algorithmic preferences and the listening patterns of the US Top 50 audience.

7.2 Promotion & Campaign Timing

- The first 30 days of chart presence are the critical intervention window. The median song disappears in 12 days; songs that survive beyond 30 days are significantly more likely to become long-tail assets. Concentrate marketing spend in this window.
- Do not rely on global Spotify popularity scores as a proxy for US chart health. With a correlation of $r = -0.094$, they measure independent things. Assign separate KPI owners for global popularity and US playlist rank.
- Invest in sync licensing and content creator campaigns for songs in their first 30 days. Viral TikTok or YouTube usage in this window is the most reliable indicator of a song transitioning from short-stay to long-tail.

7.3 Artist Development

- New Atlantic artists should target the Morgan Wallen Hit Concentration model first: build one or two inescapable hits with sustained promotional infrastructure rather than spreading investment across an album.
- Once an artist achieves Hit Concentration status (consistent top-20 rank with 5+ days average longevity), transition investment toward catalog accumulation, this is the pathway to Taylor Swift-level playlist dominance over a multi-year arc.
- Monitor the Dominance Index quarterly. Any Atlantic artist crossing 2% of playlist slots should receive dedicated catalog re-promotion to drive that figure higher.

7.4 Competitive Intelligence

- The top 3 artists (Taylor Swift, Zach Bryan, Morgan Wallen) collectively hold ~20% of playlist slots, leaving 80% contested by 293 other artists. This concentration means Atlantic artists compete intensely for a fragmented remainder.
- Sabrina Carpenter's rise to #4 dominance (967 appearances, 3.48%) demonstrates that new artists can break into the top tier within a single calendar year with the right singles and crossover strategy, this should serve as an internal benchmark.

8. Conclusion

This analysis of 27,752 daily US Spotify Top 50 playlist snapshots across 558 days reveals a nuanced picture of chart dynamics that challenges several industry assumptions. Most notably, popularity score and chart rank are nearly uncorrelated ($r = -0.094$), indicating that US domestic chart momentum operates independently of global popularity metrics, a finding with direct implications for how Atlantic's analytics teams should structure their KPI frameworks.

Artist dominance is highly concentrated: three artists account for 20% of all playlist slots, operating via three distinct strategies catalog breadth, album cut depth, and hit concentration. Each strategy is viable, but the entry points, timelines, and investment requirements differ substantially, and Atlantic should calibrate its artist development roadmaps accordingly.

The explicit content analysis reveals a clear business opportunity: clean songs chart an average of 28 days longer than explicit tracks, a 20.7% longevity advantage achievable simply by releasing a clean version alongside the standard release. This is among the highest-ROI, lowest-effort recommendations emerging from this analysis.

Finally, the chart longevity distribution confirms a universal truth in hit music: half of all songs disappear within 12 days. The first 30 days of chart presence are the critical window. Atlantic should concentrate promotional energy, sync outreach, and creator partnerships in this window to maximize the probability of crossing the threshold from short-stay to long-tail asset.

The full interactive analytics dashboard supporting this research paper is publicly available at:

spotify-us-top50.streamlit.app

— *End of Report* —